Mathematics – Arithmetic

1 Numbers and Basic Operations

1.1 Numbers: The Building Blocks of Mathematics
1.2 The Number Line: Putting Numbers in Their Place
1.3 Negative Numbers
1.4 Number Sets: Basics
1.5 Basic Operations
• Addition
• Subtraction
• Addition and Subtraction: Signed Integers
• Multiplication
• Division
• Multiplication and Division: Signed Integers
• Additive and Multiplicative Identities
1.6 Order of Operations
1.7 Absolute Value

2 Basic Numeracy and Number Theory

2.1 Divisibility
2.2 Factors
2.3 Multiples
2.4 Prime Numbers and Composite Numbers
2.5 Product of Primes and Prime Factorization
2.6 Least/Lowest Common Multiple (LCM)
2.7 Highest/Greatest Common Factor/Divisor (HCF, GCF, HCD)

3 Fractions: Parts of a Whole

3.1 Introduction to Fractions
3.2 Reducing Fractions
3.3 Equivalent Fractions
3.4 Division by Zero
• Trivia/Extra: Division by Zero
3.5 Arithmetic on Fractions
• Addition and Subtraction
• Multiplication and Division
3.6 Signs of Fractions
3.7 Reciprocals

4 Mixed Numbers and Improper Fractions: Wholes and Parts

4.1 Definitions
• Mixed Numbers
• Improper Fractions

4.2 Arithmetic on Mixed Numbers and Improper Fractions
• Addition and Subtraction
• Multiplication and Division
4.3 Conversion Between Mixed Numbers and Improper Fractions
4.4 Relationships between Fractions: Size Comparison and Parts of a Whole
4.5 Mixed Measures

5 Decimals
5.1 Place Values
5.2 Arithmetic with Decimals
5.3 Decimals and Fractions: Two Sides of the Same Coin
5.4 Repeating and Terminating Decimals
5.5 Rounding
5.6 Percentages
• Percentages as Fractions and Decimals
• Conversion Between Percentages, Fractions, and Decimals
• Percent of a Number
• Percentage Increase and Decrease (Practical Calculations)

6 Ratios and Proportions
6.1 Unit Price
6.2 Unit Multipliers / Conversion Factors

7 Exponents
7.1 Rules of Exponents
• Addition and Subtraction
• Multiplication and Division: Product and Quotient Rules of Exponents
• Power Rule of Exponents
7.2 Negative and Zero Exponents

8 Radicals
8.1 Rules of Radicals
• Addition/Subtraction
• Multiplication and Division: Product and Quotient Rules of Radicals
8.2 Radical Expressions

9 Scientific Notation
9.1 Powers of 10
9.2 Scientific Notation
9.3 Multiplication/Division
9.4 Estimation

10 Basic Measurement
10.1 Metric System and Imperial System
10.2 Length: Imperial vs Metric
10.3 Mass: Imperial vs Metric
10.4 Temperature: The Three Scales
10.5 Time: The Universal Constant